Project: Worked examples
Title: Fatigue damage (Palmgren–Miner)
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Cumulative fatigue damage of a stress-range spectrum against an S–N curve of one slope through a reference point, with an optional endurance limit below which a range does no damage; the spectrum repetitions to failure.
N_i = N_ref·(S_ref/S_i)^m, unbounded when S_i ≤ S_e. d_i = n_i/N_i. D = Σ d_i; repetitions to failure = 1/D. The damage passes when D ≤ 1.
Ref: BS EN 1993-1-9 §8 — Fatigue resistance (S–N curves by detail category)
Ref: Shigley's Mechanical Engineering Design §6-15 — Cumulative fatigue damage

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
S_ref	Reference stress range (detail category or material fatigue strength)	150	MPa	spec: SPEC-MECH-03 design criteria	1 … 2000 MPa
N_ref	Cycles at the reference stress range	1000000		spec: SPEC-MECH-03 design criteria	10000 … 1000000000
m	S–N slope	5		spec: SPEC-MECH-03 design criteria	2 … 15
S_e	Endurance limit (ranges at or below it do no damage)	60	MPa	spec: SPEC-MECH-03 design criteria	0.1 … 2000 MPa
blocks	Stress-range blocks	3 row(s)		spec: LOAD-SCHEDULE-01	any
blocks[1]		S = 120 MPa; n = 2000000			
blocks[2]		S = 90 MPa; n = 10000000			
blocks[3]		S = 50 MPa; n = 10000000			

3. Assumptions

A1. Damage accumulates linearly and independently of the order of the blocks. (Palmgren–Miner rule.)
A2. One S–N slope over the whole range; no second slope below a knee. (Simplified curve; a two-slope curve is entered as two sheets or a lower slope.)

4. Working

[5] N_1 = N_ref · (S_ref / S_1)^m
= 1000000 × (150 MPa / 120 MPa)^5
= 3051800 BS EN 1993-1-9 §8

Life at block 1

[6] d_1 = n_1 / N_1
= 2000000 / 3051800
= 0.65536

Damage of block 1

[7] N_2 = N_ref · (S_ref / S_2)^m
= 1000000 × (150 MPa / 90 MPa)^5
= 12860000 BS EN 1993-1-9 §8

Life at block 2

[8] d_2 = n_2 / N_2
= 10000000 / 12860000
= 0.7776

Damage of block 2

[9] N_3 = S_3 ≤ S_e
= 50 MPa ≤ 60 MPa
= "unbounded"

Life at block 3: below the endurance limit

[10] d_3 = n_3 / N_3
= 10000000 / ∞
= 0

Damage of block 3

[11] D = ⊠ d_i
= 0.65536 + 0.7776 + 0
= 1.433 Shigley §6-15

Total damage (Palmgren–Miner)

[12] R_f = 1 / D
= 1 / 1.433
= 0.69786

Spectrum repetitions to failure

5. Results

SYMBOL	VALUE	DESCRIPTION
D	1.433	Total damage
R_f	0.69786	Repetitions to failure
N_1	3051800	Life at block 1
d_1	0.65536	Damage of block 1
N_2	12860000	Life at block 2
d_2	0.7776	Damage of block 2
N_3	"unbounded"	Life at block 3
d_3	0	Damage of block 3

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Cumulative damage D ≤ 1	D = 1.433	D ≤ 1	1.43	Shigley §6-15

7. Applicability

No warnings.

8. Verification

Template mech.fatigue-miner@1.0.0 Content hash 9a0ade9f4660b0be Engine calc-engine/1.0.0
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